

Question 1

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions like humans. To develop an AI system, four different approaches are implemented: Acting Humanly, Thinking Humanly, Thinking Rationally, and Acting Rationally.

- a) Briefly explain the **FOUR (4)** approaches. (8 marks)

- **Acting Humanly**
 - Known as the Turing Test approach.
 - AI systems developed with this approach are capable of acting like human.
- **Thinking Humanly**
 - Known as the Cognitive Modelling approach.
 - AI systems developed with this approach are capable of thinking like human.
- **Thinking Rationally**
 - Known as the "Laws of Thought" approach.
 - AI systems developed with this approach are capable of thinking rationally.
- **Acting Rationally**
 - Known as the Rational Agent approach.
 - AI systems developed with this approach are capable of acting rationally.

- b) Identify another name for the Turing Test. (1 mark)

imitation game

- c) To pass the Total Turing Test, a computer must possess certain capabilities. List all **SIX (6)** capabilities and explain their purposes. (12 marks)

- To communicate in human languages (Natural language processing)
- To store what it knows or hears (Knowledge representation)
- To do reasoning and to draw new conclusions (Automated reasoning)
- To adapt to new circumstances and handle uncertainties (Machine learning)
- To derive visual information from its percepts (Computer vision)
- To manipulate objects and move about (Robotics)

- d) At any given time, rationality depends on four criteria to develop a rational agent. Briefly explain the **FOUR** (4) criteria. (4 marks)

- The **performance measure** that defines the criterion of success.
- The agent's **prior knowledge** of the environment.
- The **actions** that the agent can **perform**.
- The agent's **percept sequence** to date.

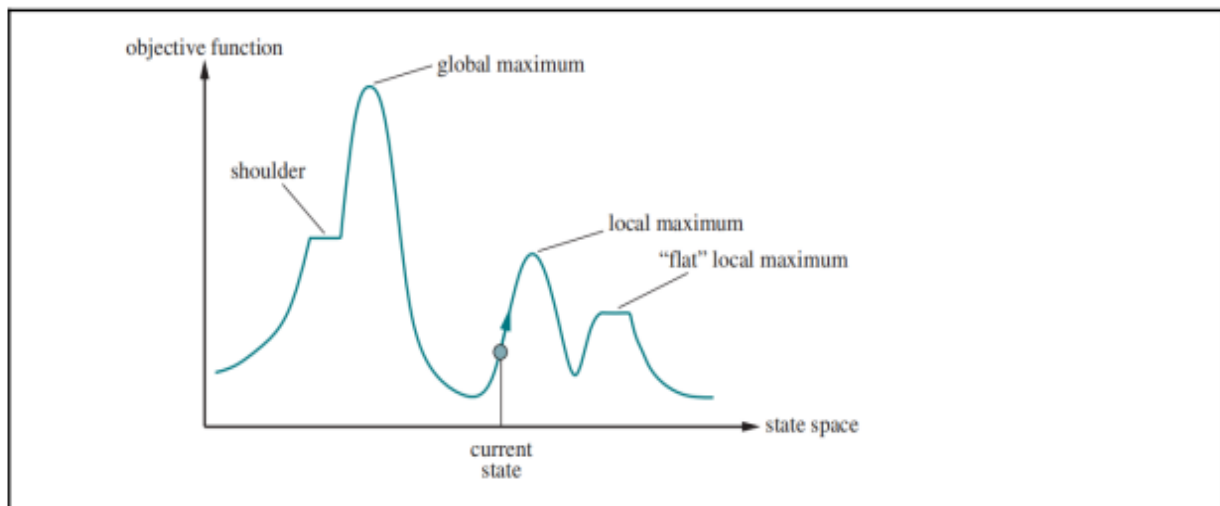
Question 2

Hill climbing is an optimisation algorithm used to identify the best possible solution to a problem. It is commonly applied in AI and machine learning for local search optimisation.

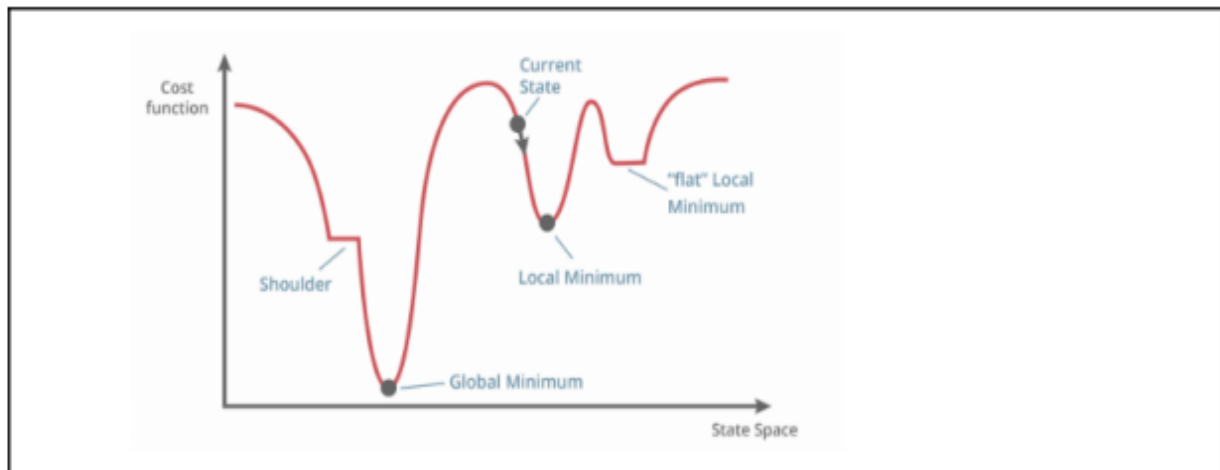
- a) Hill climbing is also known as a greedy approach or greedy local search. Briefly explain why it is referred to in this way. (2 marks)

Only consider a good neighbouring state without thinking ahead

- b) Draw and label a state-space landscape in which its elevation corresponds to an objective function. (7 marks)



- c) Draw and label a state-space landscape in which its elevation corresponds to a cost function. (7 marks)



- d) Hill climbing search often gets stuck at a sub-optimal solution. Analyse the **THREE (3)** causes.

(9 marks)

Local maxima

- A peak that is higher than each of its neighbouring states but lower than the global maximum.
- Once arrived at the local maxima, it will be stuck with nowhere else to go.

Ridges

- A sequence of local maxima that is very difficult for greedy algorithms to navigate.
- From each local maximum, all the available actions point downhill.

Plateaus

- A flat area of the state-space landscape, a flat local maximum, and a shoulder.
- Search can get lost wandering on the plateau.

Question 3

The Transportation Unit at TAR UMT has developed an AI-powered campus bus routing system designed to determine the shortest route from Station A to Station I. Figure 1 presents a directed graph consisting of nine campus bus stops, where the values indicate the travel distances (in kilometres) between each stop. Table 1 displays the heuristic cost of each stop relative to Station I, helping the system estimate the most efficient route.

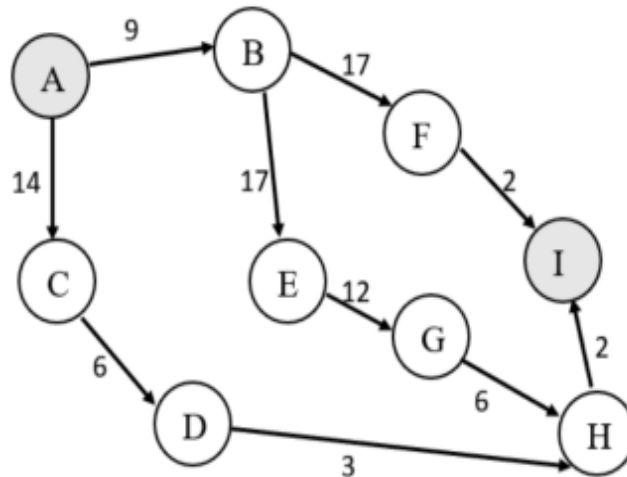


Figure 1. Search Graph

Table 1: Heuristic Cost of Each Station

Station	A	B	C	D	E	F	G	H	I
Euclidean distance	9	8	10	8	7	4	2	4	1

- a) Analyse the state space, initial state, goal state, action, transition model, and the cost function for the problem above. (6 marks)

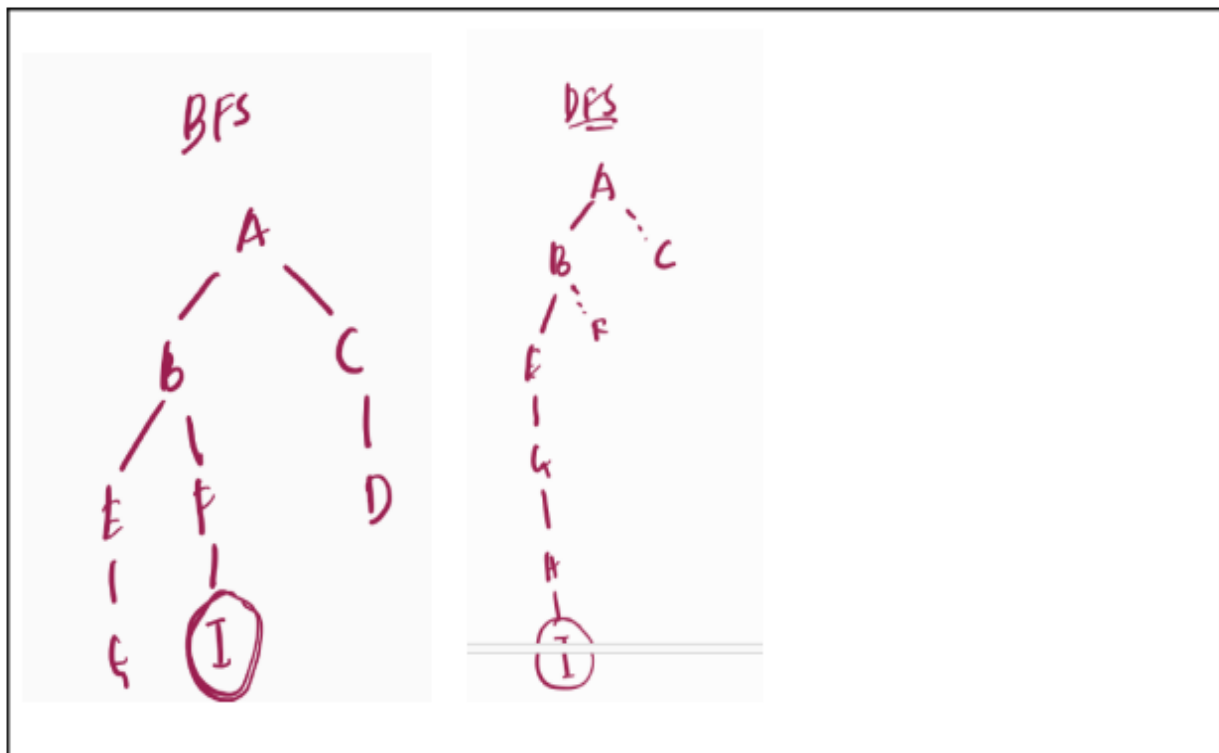
States space	The set of nodes in the search graph (A, B, C, ..., H)
Initial state	Starting node (A)
Goal state	Target node (I)
Action	Possible moves from one node to another based on the graph's edges
Transition model	Defines how each action moves from one node to another and

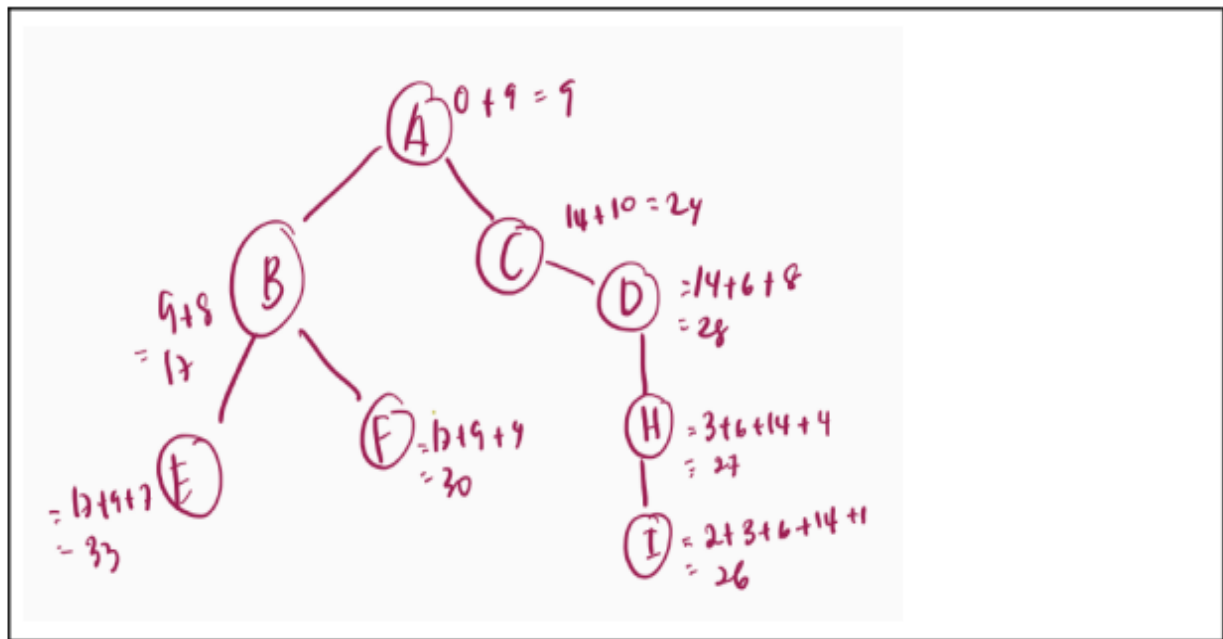
	updates the path cost.
Cost function	The evaluation function for A* is $f(n) = g(n) + h(n)$, where: • $g(n)$ = actual cost from the start node to node n • $h(n)$ = estimated heuristic cost from n to the goal

- b) Based on the information provided in Figure 1 and Table 1, illustrate the resulting search tree for Breadth-First Search, Depth-First Search, and A* search in alphabetical order.

(4 + 4 + 8

marks)





- c) Identify the solution returned by Breadth-First Search, Depth-First Search, and A* search in Question 3 b). (3 marks)

- BFS: A, B, F, I
- DFS: A, B, E, G, H, I
- A* : A, C, D, H, I

Question 4

- a) A Sudoku is a popular puzzle game based on logical number placement. It has introduced millions to Constraint Satisfaction Problems (CSP), even if they do not realise it. A classical 6×6 Sudoku consists of 36 squares, some of which are initially filled with digits from 1 to 6. The objective is to fill the grid such that: Each row, column, and each classical 2×3 sub-grid contains all digits from 1 to 6, without repetition.
- (i) Formulate the 6×6 Sudoku as a CSP and identify the variables, domains for each variable, and the constraints using the **Alldiff** global constraint. (11 marks)

Variable, $X = \{A1, \dots A6, B1, \dots B6, C1, \dots C6, D1, \dots D6, E1, \dots E6, F1, \dots F6\}$

Domain of empty squares, $D_i = \{1, 2, 3, 4, 5, 6\}$.

C =

row

Alldiff (A1, A2, A3, A4, A5, A6)

Alldiff (B1, B2, B3, B4, B5, B6)

Alldiff (C1, C2, C3, C4, C5, C6)

Alldiff (D1, D2, D3, D4, D5, D6)

Alldiff (E1, E2, E3, E4, E5, E6)

Alldiff (F1, F2, F3, F4, F5, F6)

(3%)

subgrids

Alldiff (A1, A2, A3, B1, B2, B3)

Alldiff (A4, A5, A6, B4, B5, B6)

Alldiff (C1, C2, C3, D1, D2, D3)

Alldiff (C4, C5, C6, D4, D5, D6)

Alldiff (E1, E2, E3, F1, F2, F3)

Alldiff (E4, E5, E6, F4, F5, F6)

(3%)

Column

Alldiff (A1, B1, C1, D1, E1, F1)

Alldiff (A2, B2, C2, D2, E2, F2)

Alldiff (A3, B3, C3, D3, E3, F3)

Alldiff (A4, B4, C4, D4, E4, F4)

Alldiff (A5, B5, C5, D5, E5, F5)

Alldiff (A6, B6, C6, D6, E6, F6)

(3%)

(ii) A sample Sudoku puzzle is provided in Figure 2.

	1	2	3	4	5	6
A	3	1				2
B	4	6		1		
C				5		6
D	6				2	3
E			5			
F	2				6	4

Figure 2. Sudoku Puzzle

Suppose **Alldiff** constraints have been expanded into binary constraints so that the AC-3 algorithm (i.e., the most popular algorithm for enforcing arc consistency) can be applied directly. Explain how the following variables are made arc-consistent step-by-step.

- A4
 - C2
 - E6
- marks)

(9

- A4 = {4,6}

- $C2 = \{2,3,4\}$
- $E6 = \{1\}$

However, the Sudoku is unsolvable.

- b) An Evolutionary Algorithm (EA) is a type of optimisation algorithm inspired by the process of natural evolution. It belongs to a broader class of algorithms known as evolutionary computation. These algorithms work by iteratively improving a population of candidate solutions using mechanisms that mimic biological evolution. Briefly explain the **FIVE (5)** key concepts in EA.

(5 marks)

Initialisation/Population	A group of potential solutions to the problem
Fitness Function	A function that evaluates how "good" or "fit" a solution is
Selection	The process of choosing the best individuals from the population to be parents.
Crossover (Recombination)	Combining parts of two or more parents to produce offspring
Mutation	Randomly changing parts of a solution to introduce diversity.

Goal = 1000 0000

Initial = 0000 0000 (7), 1010 0101 (4), 1001 1001(5)

Parent 1 = 0000 0000

Parent 2 = 1001 1001

Children 1 = 0000 1101

Children 2 = 1001 0000

